Initial Setup & Source Code

Before compiling, we will gather all the required compilers, multi-arch development headers, and the Mesa source code tree.

1. Prepare the Trixie Multi-Arch Environment

   Installs 64-bit tools, 32-bit tools, and WSL-specific headers

   Enable the `i386` architecture in Debian, then download every compiler and development header required for both architectures.

```bash
sudo dpkg --add-architecture i386
sudo apt update
sudo apt build-dep mesa
sudo apt install git meson ninja-build libdirectx-headers-dev dpkg-dev gcc-multilib g++-multilib pkg-config:i386 libx11-dev:i386 libdrm-dev:i386
```

2. Clone the Official Mesa Repository

   Fetch the complete Mesa source tree directly from Freedesktop's GitLab repository.

```bash
git clone https://gitlab.freedesktop.org/mesa/mesa.git
cd mesa
```

3. Checkout the Latest 26.1 Release

   Automatically targets the highest point release

   Query the repository tags, filter for the `26.1.x` branch, sort them, and check out the latest stable point release.

```bash
git checkout $(git tag | grep -E '^mesa-26\.1\.[0-9]+$' | sort -V | tail -n 1)
```

Phase 1: The 64-bit (amd64) Build (`mesa-wsl2`)

This phase compiles your primary 64-bit graphics stack, enabling Direct3D 12, OSMesa, Gallium Nine, and VA-API acceleration.

1. Configure the 64-bit Meson Build

   Activates Gallium Nine, OSMesa, and VA-API

   Configure Meson to target Debian's native multi-arch path for 64-bit libraries.

```bash
export DEB_HOST_MULTIARCH=$(dpkg-architecture -qDEB_HOST_MULTIARCH)

meson setup builddir64 \
  -Dprefix=/usr \
  -Dlibdir=lib/$DEB_HOST_MULTIARCH \
  -Dgallium-drivers=d3d12,swrast \
  -Dvulkan-drivers=microsoft-experimental \
  -Dgallium-nine=true \
  -Dosmesa=true \
  -Dgallium-va=enabled \
  -Dplatforms=wayland,x11 \
  -Dbuildtype=release \
  -Dstrip=true
```

2. Compile and Stage the 64-bit Package

   Compile the source code using Ninja and stage the output in a temporary folder.

```bash
mkdir -p /tmp/mesa-wsl-amd64
DESTDIR=/tmp/mesa-wsl-amd64 ninja -C builddir64 install
```

3. Create the 64-bit Control File

   Uses mesa-wsl2 as the primary package name

   Draft the metadata that informs APT that `mesa-wsl2` replaces Debian's stock Mesa libraries.

```bash
mkdir -p /tmp/mesa-wsl-amd64/DEBIAN
cat << 'EOF' > /tmp/mesa-wsl-amd64/DEBIAN/control
Package: mesa-wsl2
Version: 26.1.99-1
Architecture: amd64
Maintainer: Phil
Section: graphics
Priority: optional
Provides: libd3dadapter9-mesa, libegl-mesa0, libgl1-mesa-dri, libgl1-mesa-dev, libgles2-mesa-dev, libglx-mesa0, libosmesa6, mesa-libgallium, mesa-va-drivers, mesa-vulkan-drivers, libglapi-mesa, vulkan-icd
Replaces: libd3dadapter9-mesa, libegl-mesa0, libgl1-mesa-dri, libgl1-mesa-dev, libgles2-mesa-dev, libglx-mesa0, libosmesa6, mesa-libgallium, mesa-va-drivers, mesa-vulkan-drivers, libglapi-mesa
Conflicts: libd3dadapter9-mesa, libegl-mesa0, libgl1-mesa-dri, libgl1-mesa-dev, libgles2-mesa-dev, libglx-mesa0, libosmesa6, mesa-libgallium, mesa-va-drivers, mesa-vulkan-drivers, libglapi-mesa
Description: Custom 64-bit Mesa 26.1 build optimised for WSL2
 Provides core D3D12 drivers, OSMesa, and Gallium Nine support
 for nested Wayland environments.
EOF
```

4. Build the 64-bit Package

   Package the staging directory into your primary `.deb` file.

```bash
dpkg-deb --build /tmp/mesa-wsl-amd64 mesa-wsl2_amd64.deb
```

Phase 2: The 32-bit (i386) Build (`mesa-wsl2-32bit`)

To get a 32-bit `libd3dadapter9-mesa` adapter for 32-bit Wine applications, we will cross-compile the Mesa tree targeting the `i386` architecture.

1. Create a Meson Cross-File

   Tells Meson to use 32-bit compilation flags

   Write a cross-compilation file so Meson targets 32-bit compilation.

```bash
cat << 'EOF' > /tmp/mesa-i386-cross.txt
[binaries]
c = 'gcc'
cpp = 'g++'
pkgconfig = 'i686-linux-gnu-pkg-config'

[built-in options]
c_args = ['-m32']
c_link_args = ['-m32']
cpp_args = ['-m32']
cpp_link_args = ['-m32']

[host_machine]
system = 'linux'
cpu_family = 'x86'
cpu = 'i686'
endian = 'little'
EOF
```

2. Configure and Compile the 32-bit Build

   Set up the `i386` build directory using the cross-file and stage the compiled files.

```bash
meson setup builddir32 \
  --cross-file /tmp/mesa-i386-cross.txt \
  -Dprefix=/usr \
  -Dlibdir=lib/i386-linux-gnu \
  -Dgallium-drivers=d3d12,swrast \
  -Dgallium-nine=true \
  -Dplatforms=x11 \
  -Dbuildtype=release \
  -Dstrip=true

mkdir -p /tmp/mesa-wsl-i386
DESTDIR=/tmp/mesa-wsl-i386 ninja -C builddir32 install
```

3. Create the 32-bit Control File

   Uses mesa-wsl2-32bit as the 32-bit package name

   Draft the metadata for your 32-bit package.

```bash
mkdir -p /tmp/mesa-wsl-i386/DEBIAN
cat << 'EOF' > /tmp/mesa-wsl-i386/DEBIAN/control
Package: mesa-wsl2-32bit
Version: 26.1.99-1
Architecture: i386
Maintainer: Phil
Section: graphics
Priority: optional
Provides: libd3dadapter9-mesa
Replaces: libd3dadapter9-mesa
Conflicts: libd3dadapter9-mesa
Description: Custom 32-bit Mesa 26.1 build optimised for WSL2
 Provides 32-bit Gallium Nine support for Wine applications.
EOF
```

4. Build and Install Both Packages

   Build the 32-bit `.deb` file and install both packages together using `apt`.

```bash
dpkg-deb --build /tmp/mesa-wsl-i386 mesa-wsl2-32bit_i386.deb
sudo apt install ./mesa-wsl2_amd64.deb ./mesa-wsl2-32bit_i386.deb
```